

CSC240DynamicArrayQueue/evacuateDriver.cpp

```
1  /*
2   * evacuateDriver.cpp
3   *
4   * Created on: Mar 4, 2020
5   * Author: Ivan
6   */
7  #include <iostream>
8  #include <random>
9  #include <ctime>
10 #include "QueType.h"
11 using namespace std;
12
13 int main(){
14     const int NUM_CUSTOMERS = 50;
15     const int NUM_CHARACTERS = 26;
16     QueType line(NUM_CUSTOMERS);
17     int numEvacuate;
18     int numRemoved = 0;
19     char evacuee;
20     char alphabet[] = {'a', 'b', 'c', 'd', 'e', 'f', 'g', 'h', 'i', 'j', 'k', 'l',
    'm', 'n', 'o',
21                       'p', 'q', 'r', 's', 't', 'u', 'v', 'w', 'x', 'y', 'z'};
22     srand(time(0));
23
24     QueType testLine(2);
25     testLine.Enqueue('a');
26     cout << testLine << endl;
27     testLine.Enqueue('b');
28     cout << testLine << endl;
29
30     for(int i = 0; i < NUM_CUSTOMERS; i++){
31         char randChar;
32         randChar = alphabet[rand() % NUM_CHARACTERS];
33         //cout << randChar << endl;
34         line.Enqueue(randChar);
35     }
36
37     //Before evacuate
38     cout << "Before evacuation:\n" << line << endl;
39
40     cout << "How many characters should evacuate from the rear?";
41     cin >> numEvacuate;
42     line.evacuate(numEvacuate);
43     numRemoved += numEvacuate;
44
45     //after evacuate
46     cout << "After evacuation:\n" << line << endl;
```

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47
48     cout << "Rear item in line dequeued: " << line.DequeueRear() << endl;
49     cout << line << endl;
50     numRemoved++;
51
52     line.Dequeue(evacuee);
53     cout << "Front item in line dequeued: " << evacuee << endl;
54     cout << line << endl;
55     numRemoved++;
56
57     line.evacuate(NUM_CUSTOMERS - numRemoved - 1);
58     cout << "Evacuate all but one:\n" << line << endl;
59     numRemoved += (NUM_CUSTOMERS - numRemoved - 1);
60
61     line.evacuate(1);
62     cout << "Evacuate last one:\n" << line << endl;
63
64
65     return 0;
66 }
67
68
```